

Section: **3.1 – STRENGTH OF METALLIC PARTS**
Chapter: **3185**
Title: **EFFECTIVE SKIN WIDTH**
Revision: **0.3**
Date: **22/06/2020**

1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

This chapter defines the methodology to use for computing the effective skin width in metallic stiffened panels subjected to compression load, when the skin is allowed to buckle between stringers (local buckling). The effective width approach traditionally supports the stress analysis of post-buckled stiffened panels, subjected to compression loads, up to failure.

Extract from Bruhn's C7.11... *The major structure of aerospace vehicles, such as the wing or body, involves a sheet covering which is strengthened by attached or integral fabricated stiffeners such angles, zeels, tees, etc. Since the sheet and stiffeners much deform together, the sheet will therefore carry compressive load and to neglect this load carrying capacity of the sheet would be too conservative in aerospace structural design where weight saving is very important.*

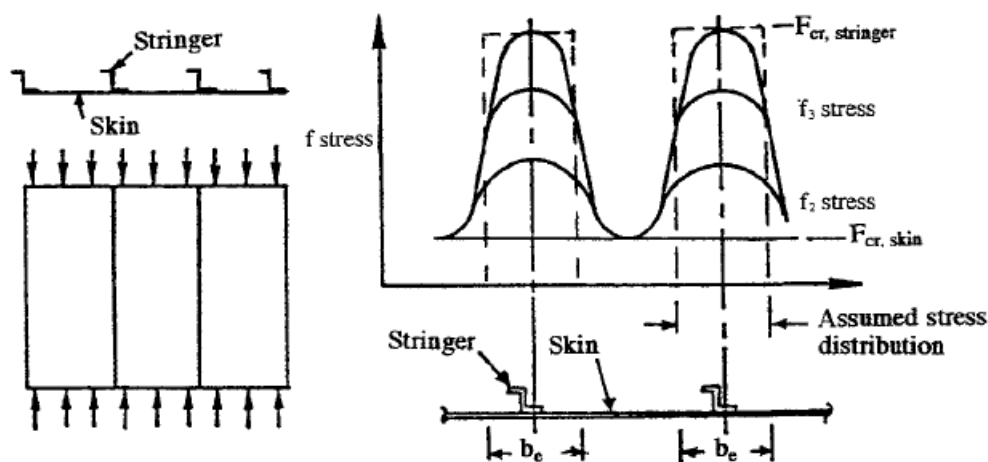


Figure 1. The effective width of a skin-stringer panel (Niu)

In tension, on longitudinal straight stringers, the whole skin width shall be considered effective for carrying axial loads. However, for the skin on curved frames, when subjected to hoop loads (eg. due to cabin pressurization), the calculation of the effective width is a more complex problem. As discussed by Schagerl (Ref. 3): *The skin of a curved aircraft panel flattens, if it is subjected to circumferential tension, and bulges, if it is subjected to compression. These deformations distort the hoop stress distribution, which further leads to a reduced load-carrying width of the skin in circumferential direction.*

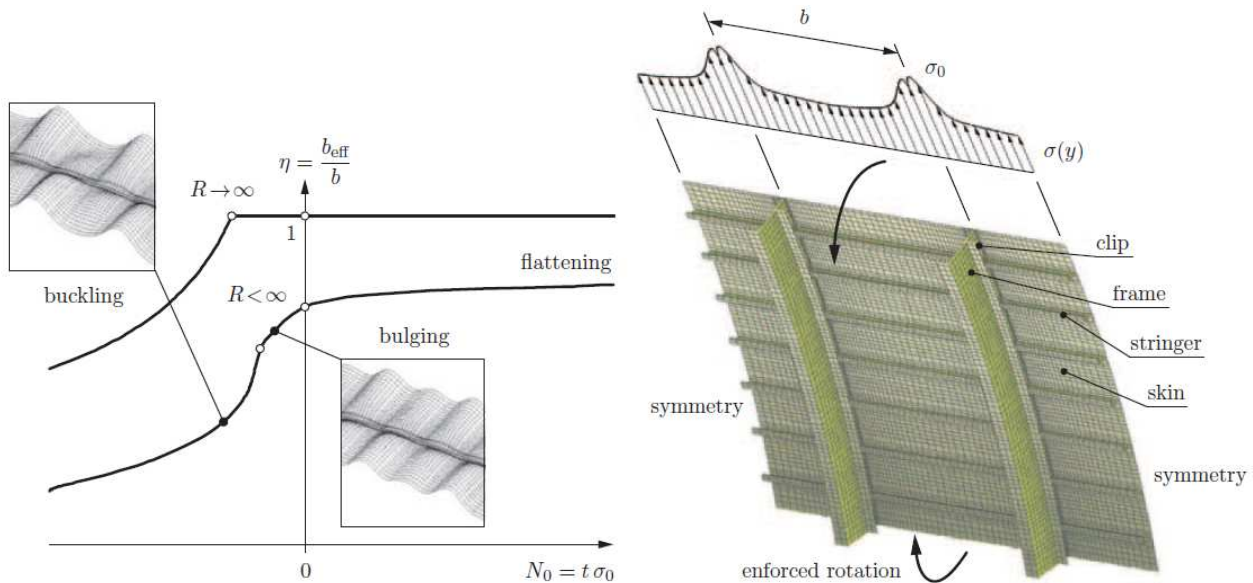


Figure 2. Flattening (tension) and bulging (compression) in curved skins at frame supports

Remark: The effective width problem of curved skins on frames shall not be treated in present revision of this chapter.

2 METHOD DESCRIPTION

2.1 EFFECTIVE WIDTH OF SKIN ON STRAIGHT STRINGERS

The compression buckling stress of a supported plate/sheet of width b is:

$$\sigma_b = K_c \cdot E \cdot (t/b)^2 \quad (\text{Niu 11.2.2, Bruhn C7.13-14})$$

In a stiffened panel, the supports of the skin, the stringers, shall be able to sustain more compression stress beyond the skin local buckling. The sheet in the buckled state has the ability to approximately hold this stress as the stiffener continues to take load until it reaches the stiffener crippling stress. The effective skin width is defined to set a cross section area at the same stress as the stiffener equivalent to this stringer and the buckled skin sheet. The classical expression for this is:

$$w = t \cdot (K_c \cdot E / F_{st})^{0.5} \quad (\text{Niu 14.2.1})$$

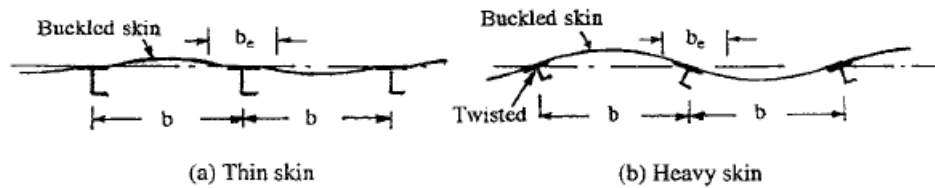


Figure 3. Thin and heavy skins (Niu)

When K_c is made equal to 3.62 (the “heavy skin” mode in Niu’s book), the familiar expression results:

$$w = t \cdot (3.6 \cdot E / F_{st})^{0.5} = 1.9 \cdot t \cdot (E / F_{st})^{0.5} \quad (\text{Bruhn C7.16})$$

Greater values of K_c are accepted by Niu from b/t ratios of 40 up to the “fixed edges” condition at b/t equal to 110 of “thin skins” (Niu’s Fig. 14.2.3). This potential improvement, which does not explicitly account for the actual stiffness of the stringer, shall not be applied. On the contrary... Newell suggested to change the coefficient 1.9 to 1.7 in configuration with light stringers. This later form has been widely used in the industry.

$$w = 1.7 \cdot t \cdot (E / F_{st})^{0.5}$$

For panels with different thickness at each side of the stringer, it will be practical to use the semi-effective width instead:

$$w_i = 0.85 \cdot t \cdot (E / F_{st})^{0.5}$$

It is remarked that the E in the formula is the Young’s modulus of the material, even when the stress of the stringer exceeds the yield stress (see discussion in Bruhn C7.12). Niu suggests to apply a plasticity correction factor to the Young modulus only for b/t ratios below 40. This is done in order to account for the reduction of the initial buckling stress due to plastic effects in narrow plates (with typical aluminum alloys used in aerospace, it is easy to check, with first formula in this page, that this approach is conservative).

Note that, as presented above, the effective width is function of the applied loads (the stress at the stringer/skin junction). It is reasonable to simplify this expression by conservatively use the value at the stiffener failure stress, i.e. at the stiffener crippling stress, no matter what is the applied compression load. In this case the section inertia would be underestimated below the stringer crippling load, which will affect the calculation of stress distribution in the section, or the column buckling load.

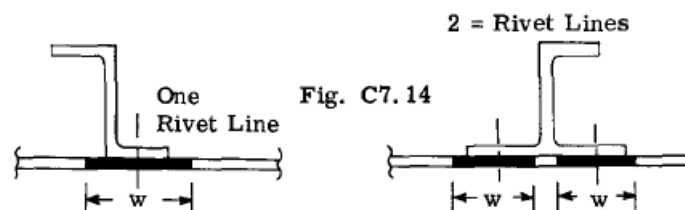


Figure 4. Effective width for sheet-stiffener units fastened by single attachment lines

Previous formulas are felt as appropriate for general design purposes. Literature presents also alternatives for other configurations, when the stiffener can be expected to provide support to the skin beyond the simple-support condition (see for example suggestions in Bruhn C7.12 or Niu 14.2, for integral stringers or staggered rivets rows).

Effective width corrected with plasticity

As discussed in previous section, the effective width shall be calculated with the Young modulus corrected by plasticity only in narrow plates (b/t ratios below 40). The plasticity correction factor η applicable ($\eta \cdot E$ replacing E in previous formulas) shall be that corresponding to buckling of flat plates in compression. Some classical expressions available for this problem are:

- Niu, section 11.2 (B) $\eta = (E_t/E)^{0.5}$
- Bruhn, Fig. C5.8 & NACA TN 3781 $\eta = 0.5 \cdot (E_s/E) \cdot \{1 + 0.5 \cdot [1 + (3 \cdot E_t/E_s)]^{0.5}\} \cdot (1 - \nu_e^2)/(1 - \nu^2)$
with $\nu = \nu_p - (E_s/E) \cdot (\nu_p - \nu_e)$

Tangent and secant moduli in expressions above are function of the stress level in the material. In this case, in order to be consistent with the general approach and Niu's discussion, E_t and E_s values for obtaining η shall be computed at the local buckling stress level of the skin panel.

Effective width corrected due to different stringer material

The sheet directly adjacent to the stiffener attachment line must undergo the same strain as the stiffener. In case the skin and the stringer are made of different materials, the correction for this condition can be made by multiplying previous value by F_{sh}/F_{st} :

$$w' = w \cdot (F_{sh} / F_{st}) \quad (\text{Bruhn, formula C7.18})$$

... where F_{sh} is the stress at the skin sheet for the strain level seen by the stringer.

Effective width corrected with inter-rivet buckling in skin

When the sheet between rivets buckles before the crippling stress of the stiffener is reached, the sheet in the buckled state has the ability to approximately hold this stress as the stiffener continues to take load until it reaches the stiffener crippling stress. This buckling sheet strength can be taken advantage of by reducing the effective sheet area. Thus effective width equals,

$$w'' = w' \cdot (F_{ir} / F_{st}) \quad (\text{Bruhn, formula C7.23})$$

2.2 EFFECTIVE WIDTH OF SKIN ON CURVED FRAMES

Reserved for future use.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
b	mm	Skin width, distance between stiffeners
E	MPa	Young's modulus of the material
E_s	MPa	Secant modulus of the material
E_t	MPa	Tangent modulus of the material
F_{sh}	MPa	Compression stress in the skin at the strain level of the stiffener
F_{st}	MPa	Compression stress of the stiffener
F_{ir}	MPa	Inter-rivet buckling stress
K_c	-	Coefficient in buckling formula (3.6 for simple-support conditions)
t	mm	Thickness of the skin
w_i	mm	Semi-effective skin width at one side of the stiffener
w	mm	Effective skin width
σ_b	MPa	Buckling stress of the skin in compression
η	-	Plasticity correction factor for Young's modulus in buckling formulas
ν	-	Poisson ratio of the material
ν_e	-	Poisson ratio in elastic regime, equal to 0.33 for aluminum aerospace alloys
ν_p	-	Poisson ratio in fully plastic regime, equal to 0.5 for isotropic materials

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 [E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2 [M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3 [M. Schagerl]	2011	Analytical formulas for the effective width caused by bulging and flattening of curved aircraft panels
Ref. 4 [NACA] TN 3781	1957	HANDBOOK OF STRUCTURAL STABILITY. PART I – BUCKLING OF FLAT PLATES.

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.3 22/06/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 22/06/2020	Florian Glock XX/06/2020

Table 3. *Approval signatures table for last revision of this chapter*